BOTSWANA INTERNATIONAL UNIVERSITY OF SCIENCE & TECHNOLOGY

College of Sciences

Department of Computer Science & Information Systems

COMP 302 – Data Structures & Algorithms

Module Study Guide

Academic Year 2017/2018

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MODULE DESCRIPTION

This module will introduce the principal fundamental data structures and algorithms used in computer science. Data structures will be formulated to represent information in such a way that it can be conveniently and efficiently manipulated by the algorithms that are developed. The ideas will be presented abstractly, although examples will be given in the language used in the programming workshop module.

Data structures and Algorithms is a core course for students majoring in Computer Science. The main objective of the module is to teach the student how to select and design data structures and algorithms that are appropriate for problems that they might encounter. This module is also about studying their computational complexities. This module offers the student a mixture of theoretical knowledge and practical experience. At the end students will be able to design efficient algorithms in terms of time and space requirements

Prerequisites

We assume that the student is at least vaguely familiar with a high-level programming language, such as C, C++, Python, or Java, and that he or she understands the main constructs from such a high-level language, including:

- Variables and expressions
- Methods (also known as functions or procedures)
- Decision structures (such as if-statements and switch-statements)
- Iteration structures (for-loops and while-loops)

In terms of mathematical background, we assume the student is somewhat familiar with topics from high-school mathematics.

Learning Outcomes

A student who successfully completes the module will have the ability to:

- Understand a variety of techniques for designing algorithms.
- Understand a wide variety of data structures and should be able to use them appropriately to solve problems.
- Design a variety of data structures and algorithms and should be able to use them appropriately to solve problems in scientific computing.
- Understand some fundamental algorithms used in scientific computing and solve computational problems using these algorithms.
- Develop new or re-use already existing efficient algorithms to solve problems.
- Comprehend algorithmic complexity and be able to make informed choices when doing problem solving.

MODULE STAFF

Module Coordinator

Your module coordinator will be the main contact person for the module. The coordinator will be available to assist you in anything concerning the module or refer you to an appropriate staff member who will assist you. The Coordinator for this module is:

Dr. Keletso Letsholo
Lecturer, Computer Science & Information Systems
Office: Room 31, Office Block 117
E-mail address: letsholok@biust.ac.bw

Teaching Assistants

- Ronnie Nkhori (ronnie.nkhori@studentmail.biust.ac.bw)
- Randine Tjiriange (randine.tjiriange@studentmail.biust.ac.bw)
- Thabang Ramaijane (thabang.ramaijane@studentmail.biust.ac.bw)

How to Seek Assistance

If at any point during the course of the module you need help, you are to contact the module coordinator preferably by email. The Coordinator shall assist you or direct you to the appropriate staff member.

Email will be used most for communication. It is the responsibility of all students to make sure their email is functional and they check it regularly. All students should make sure they have access to blackboard.

SYLLABUS

The following is an outline of the topics covered in this module:

Week	Learning Unit	Reading
1	**Introductions** ▪ What is Data Structures? ▪ Abstract Data Type (ADT) ▪ What is Algorithm? ▪ Properties of Algorithms ▪ The need for Data Structures ▪ Philosophy of Data Structures ▪	Chapter 1 & 2
2	**Arrays and Linked Lists** ▪ Arrays ▪ Singly-Linked Lists ▪ Doubly-Linked Lists	Chapter 3
3	**Recursion** ▪ Foundations of recursion ▪ What is recursion? ▪ Recursive Algorithm design ▪ Applications of recursion	Chapter 5
4	**Stacks and Queues** ▪ Stack ADT ▪ Queue ADT ▪ Array implementation of Queues	Chapter 6
5	**Analysis of Algorithms** ▪ Empirical analysis ▪ Theoretical analysis ▪ Algorithmic time complexity ▪ Big-Oh Notation	Chapter 4
6	**Tree Structures** ▪ What is a Tree? ▪ Tree ADT ▪ Binary Trees ▪ Binary search Tree ▪ Tree traversal algorithms	Chapter 8
7	**Mid Semester Break**	

8	**Sorting and Selection** ▪ What is sorting? ▪ Selection Sort ▪ Insertion Sort ▪ Bubble Sort	Chapter 13
9	**Sorting and Selection** ▪ Merge Sort ▪ Quick Sort ▪ Comparing Sorting Algorithms	Chapter 13
10	**Graphs** ▪ What is a Graph? ▪ The Graph ADT ▪ Data Structures for Graphs	Chapter 14
11	**Graphs** ▪ Graph Traversals - Depth-First Search - Breadth-First Search ▪ Shortest Paths - Weighted Graphs - Dijkstra's Algorithm	Chapter 14
12	**Maps and Hash Tables** ▪ The Map ADT ▪ Hash Functions	Chapter 14
13	**Revision**	

LEARNING MATERIAL

Data Structures and Algorithms in Java 6th Edition by Michael T. Goodrich (Author), Roberto Tamassia (Author), Michael H. Goldwasser (Author)

MODULE ASSESSMENT

Criteria

Continuous Assessment will contribute 50% of the total score, with the other 50% from the Final examination, to be written at the end of the semester

Continuous Assessment (50%)

The coursework is organized around one theory test and two lab assignments.

Lab Exercises: 35%

- Lab exercises (Lab 1, 2 & 3) to be assessed **(10%)**
- Group assignment 1: due in week 8 **(10%)**
- Group assignment 2: due in week 11 **(15%)**

Theory Test: 15%

- TBA

Final Examination (50%)

The final examination written at the end of the semester will contribute 50% to the final mark.

Submitting Assessments

You may also be required to sign a declaration, stating that the assignment you have submitted is truly your own work and not copied, plagiarized either from other you, textbook, encyclopedias, journals or the internet. If it is discovered that you have breached this declaration, a mark of zero shall be given.

Late Work

Late submission of work will be penalized by a reduction of 10% off the awarded mark for every working day that the assignment is handed in late. Thus if an assignment is 3 days late the awarded mark will be reduced by 30%. After three working days the assignment will not be accepted and a zero will be awarded. All lecturers and instructors will automatically enforce this procedure.

If you believe you have a valid reason for late submission you must provide documentary evidence to the module coordinator **before** you miss a test or assignment

DULY PERFORMED (DP) REQUIREMENTS

A DP certificate must be earned in order for a student to write an examination at the end of the course. The requirements for the award of a DP are:

i) Attendance at 95% of lectures, tutorials and practicals. All practicals and tutorials missed without valid reasons will count towards the 5% that a student has missed.

ii) Continuous assessment of at least 40%. The CA mark is calculated from marks allocated for performance during the course, assessed as listed above under "Continuous assessment". These will be published at the end of the module before the final examination.

A list of those students refused DP certification shall be published on or before the last day of teaching of the semester.

Leave of Absence

An attendance register will be taken for each class and to obtain your DP you will need to attend 95% of your classes.

It is, therefore, imperative to notify the **College Manager** if you know you will be absent, using the appropriate form, at least one week in advance. The College reserves the right **not** to grant Leave of Absence in advance.

If you miss class it is important that you notify the **College Manager** by filling in the appropriate form and submit it with documentary evidence of why you missed the classes, e.g. a doctor's note, a note from a clinic, or a letter from a close family member in the case of funerals.

If you do not adhere to these procedures you may be refused your DP or if you miss a test or laboratory or assignment you will receive a mark of zero.

There will not be an opportunity for student to do practical outside the scheduled practice times. If you had a valid reason and wish to do a practical on a day other than that which you have been allocated, you will need permission from the module coordinator to do so. Please note that it is your responsibility to inform the module coordinator well in time in order to arrange this.

BEHAVIOUR IN CLASS

It is expected that students will behave in a manner which does not impede the ability of other students to carry out their work in class. This includes: eating or drinking in class; any kind of noisy or disruptive behavior this includes talking in lectures; using cell phones in any class is forbidden, this includes receiving calls or sms or replying to calls or sending any kind of message; and the use of iPODs or any other listening device. No audio or visual recording of any kind is allowed in any class. If any behavior is deemed to be disruptive, the lecturer or instructor may ask you to leave the class and any cell phone or other device

confiscated. Any such behaviour will be reported and may be acted upon through the Student Code of Conduct Regulations.

LECTURES AND LAB SESSIONS

Attending every lecture is crucial to learning the material necessary to complete the assignments and practical programs. Handouts generally will not be provided in printed form. However, slides will be posted after lectures in the black board, and you are welcome to print them yourselves. Some students may seek to take another course that overlaps with the lecture time and either come late to lecture or leave early. This should be avoided, as it disrupts the lectures and distracts the rest of the class. *Please avoid interrupting the class by arriving late or leaving early*.

The weekly laboratory sessions are mandatory to attend and it is expected that you will complete the in-lab portions during the lab hours. Lab assistants will be in lab at the time to assist you and verify that you have completed the in-lab exercises. Lab assistants are expected to encourage students to use Linux computing environment present on the machines in the lab. Using a single standardized development environment will make debugging easier and ensure consistent results across machines.

SAFETY IN THE LABORATORY

This module comprises laboratory practical component. This will be carried out in the Computer labs. It is therefore essential that everyone practices proper safety precautions. Please make sure you refer to the module guidelines and laboratory handout on safety and the consequences of not following safety precautions.

ACADEMIC MISCONDUCT

A student shall be guilty of Academic Misconduct if he or she breaches academic rules or regulations, including false or fraudulent representation, cheating, plagiarism, intimidation, bribery or attempted bribery, disruption of teaching, theft, copyright misuse and intellectual property theft; possessing, using, or exchanging improperly acquired written or verbal information in the preparation of laboratory work, a report, essay, examination, or other academic assignment; unauthorised interaction with another student or students in the submission of academic requirements, false claims for work submitted by a student, alteration of any academic grade or rating so as to obtain unearned academic credit, forgery, alteration, or misuse of any University or College document relating to the academic status of a student.

PLAGIARISM

Plagiarism Is a Serious Offence; the copying of other people's work without full acknowledgement is **plagiarism** and is regarded as a very serious offence in all contexts. The University regards this as a particularly serious offence, which will be recorded on the academic record of any student found guilty of plagiarism. Such a student will either be

asked to sign an Admission of Guilt form or have the matter referred to the Student Code Committee.

Following the signing of an Admission of Guilt form, the student's name, student number and offence are published on the University Notice System and any mark for the relevant assignment is cancelled.

In serious cases, following a hearing in the Student Code Committee, the student may be excluded from the University.

There is an increasing tendency for students to find suitable internet sites and simply copy large parts of these sites as their own work. This is easily detected. While students are encouraged to use the internet, note that you are obliged to **acknowledge your sources**.